Discrete optimization

Definitions

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Optimization: principles and algorithms

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Definitions

Discrete optimization involves decision variables that must be integer
We define here some variants of discrete optimization problems.

Discrete optimization

Integer Linear Problem

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$\begin{aligned} Ax &= b \\ x &\geq 0 \\ x &\in \mathbb{Z}^n \end{aligned}$$

Mixed Integer Linear Problem

$$\min_{x \in \mathbb{R}^n, y \in \mathbb{R}^p} c_x^T x + c_y^T y$$

subject to

$$\begin{aligned} A_x x + A_y y &= b \\ x &\geq 0 \\ y &\geq 0 \\ y &\in \mathbb{Z}^p \end{aligned}$$

Binary linear optimization problem

$$\min_{x \in \mathbb{N}^n} c^T x$$

subject to

$$Ax = b$$
$$x \in \{0, 1\}^n$$

Transformation

Consider $x \in \mathbb{N}$, $x \leq u$.

$$x = \sum_{i=0}^{K-1} 2^i z_i.$$

$$K = \lceil \log_2(u+1) \rceil.$$

$$u = 5. K = 3. x = z_0 + 2z_1 + 4z_2.$$

$$\sum_{i=0}^{K-1} 2^i z_i \leq u.$$

Combinatorial optimization

$$\min f(x)$$

subject to

$$x \in \mathcal{F} \text{ a large finite set.}$$

Summary

We have defined integer linear optimization problems,
mixed integer linear optimization problems,
binary linear optimization problems, and
combinatorial optimization problems.
We have seen that any integer variable can be replaced by a set of binary variables.